

A+ - Understanding And Maintenance Of Networks

Section 1: Multiple Choice

1. What is the primary function of a router in a computer network?
 - a) Assigning IP addresses to devices
 - b) Providing wireless connectivity to devices
 - c) Forwarding data packets between networks**
 - d) Managing user authentication and access control
2. What is the purpose of DNS (Domain Name System) in a computer network?
 - a) Encrypting data transmissions for security
 - b) Assigning IP addresses to devices dynamically
 - c) Converting domain names to IP addresses**
 - d) Routing data packets between network segments
3. What type of network topology uses a centralized hub or switch to connect all devices?
 - a) Star**
 - b) Bus
 - c) Ring
 - d) Mesh
4. Which network protocol is commonly used for securely accessing and transferring files over a network?
 - a) HTTP
 - b) FTP**
 - c) SMTP
 - d) POP3

Section 2: True or False

5. A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules. **True**
6. DHCP (Dynamic Host Configuration Protocol) assigns static IP addresses to network devices automatically. **False**
7. VLANs (Virtual Local Area Networks) enable network segmentation by dividing a single physical network into multiple logical networks. **False**

Section 3: Short Answer

8. Explain the difference between a hub and a switch in a computer network.

Answer.

Hub	Switch
Works on Physical Layer (Layer 1).	Works on Data Link Layer (Layer 2).
Sends data to all connected devices	Sends data only to the target device
Does not check MAC address.	Uses MAC address to forward data.
More network collisions.	Reduces collisions.
Slower performance.	Faster and more efficient.
Less secure.	More secure.

9. Describe the process of troubleshooting network connectivity issues.

Answer :- Troubleshooting of network connectivity issue:-

- First, check the cables are properly connected.
- Then check the router lights to ensure the router is working correctly.
- Open Command Prompt and type ipconfig to check the IP address.
- If the IP address starts with 192.168.x.x, it is valid.
- If it starts with 169.x.x.x, it indicates a DHCP problem.
- Then ping the gateway using ping 8.8.8.8.
- In cmd enter:-
 - ipconfig /release
 - ipconfig /renew.
- If needed, use ipconfig /flushdns.
- Check the network adapter settings and ensure it is set to Obtain an IP address automatically.
- If still not working, go to Device Manager, uninstall the network driver, and reinstall it.
- Then check in services.msc whether the DHCP Client service is set to *Automatic* and running.
- If the problem still continues, restart the router or contact the ISP.

Section 4: Practical Application

10. Demonstrate how to configure a wireless router's security settings to enhance network security.

Answer :- To configure a wireless router's security settings:

First, connect the router to the computer and find the router's IP address (for example, 192.168.1.1). Enter this IP address in a web browser.

Next, log in using the router's username and password.

After logging in, open the Wireless or Wi-Fi Configuration page. Here, you can:

- Change the Wi-Fi name (SSID).
- Set a strong Wi-Fi password.
- Select a secure encryption type such as WPA3 or WPA2-PSK.
- Configure the internet connection type (such as PPPoE, if required by the ISP).

Finally, save the settings and restart the router to apply the changes.

Section 5: Essay

11. Discuss the importance of network documentation and provide examples of information that should be documented.

Answer :- Network documentation is important because it helps organizations manage and maintain their network effectively.

It provides clear information about how the network is designed, configured, and connected.

Importance :-

- It ensures consistency and accuracy in network management.
- It helps quickly identify the root cause of network issues.
- It supports better decision-making during upgrades or expansion.
- It improves accountability and security control.
- It reduces dependency on a single IT person.

Network documentation provides clarity, control, and continuity in managing a network.

Examples of information that should be documented in a network :-

- Cables connections and port number
- Backup & Restore process
- Firewall & security configurations
- Device Details (router, switch, etc..)
- Network drivers and configurations settings.